

NOR-TL: NORMAL-REFERENCE TEMPORAL LEARNING WITH DECOUPLED SUPERVISION FOR PAROXYSMAL AF DETECTION

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ABSTRACT

While atrial fibrillation (AF) classification from routine short clinical electrocardiogram (ECG) recordings has achieved strong performance, long-term paroxysmal-AF (PAF) monitoring in daily life requires distinguishing AF episodes from surrounding periods of normal rhythm over time. However, the substantial effort required for temporal annotation limits the amount of densely annotated data available for learning generalizable representations. To address this limitation, we propose NOR-TL, a framework that unifies large-scale ECG pretraining with limited fine-grained interval supervision. NOR-TL combines a pretrained encoder that extracts transferable normal-reference features with an adaptive stream for AF-specific modeling. NOR-TL decouples supervision through a shared temporal predictor: weak recording labels guide pooled detection via multiple instance learning (MIL), while interval annotations supervise sample-wise AF prediction. Evaluated on AFDB, LTAfDB, and CPSC2021, NOR-TL achieves the highest time-point MCC among the locally adapted baselines, reaching 98.0% temporal AUROC and 0.838 MCC on CPSC2021. These results demonstrate the effectiveness of the proposed normal-reference conditioning and decoupled supervision across annotation granularities.

Index Terms— Paroxysmal atrial fibrillation, single-lead ECG, deep learning, multi-granularity supervision

1. INTRODUCTION

Automated atrial fibrillation (AF) classification from routine short clinical electrocardiogram (ECG) recordings has achieved strong diagnostic performance [1, 2]. For long-term monitoring of paroxysmal-AF (PAF), however, a short recording provides only a snapshot of rhythm, whereas intermittent transitions between AF and normal rhythm require tracking episodes over time. The task therefore extends beyond recording-level classification to tracking intermittent AF periods of varying duration, which can be difficult to capture during long-term monitoring. Quantifying their cumulative duration enables assessment of AF burden, defined as the proportion of monitored time spent in AF [3, 4]. This need is particularly relevant to portable and home-monitoring settings that support extended ECG recording [5]. Previous work has shown that combining complementary waveform and rhythm information may improve AF detection [6]. In this context, discriminative features can capture AF-related changes as well as the normal waveform and rhythm

patterns surrounding each episode, allowing the transitions between the two states to be identified.

Learning to distinguish AF from normal rhythm across patients requires training data that capture variation in both rhythm states [7]. Available interval-annotated cohorts, however, contain recordings from relatively few patients. AFDB, for example, provides only 23 waveform records [8], which is far less than necessary. An intuitive choice is to split long recordings into many short window snapshots, which can increase the number of training examples but does not expand the patient cohort. At the same time, the large-scale public ECG datasets provide extensive samples across patients and recording conditions [9, 10], and previous studies have used pretraining and contrastive learning to develop ECG representations from large or multi-source collections [11, 12, 13]. As expected, the labels in large clinical ECG collections are typically available only at the recording level, whereas sample-wise AF prediction requires annotations of episode extent.

Multiple instance learning is capable of linking local evidence to recording-level labels without treating the coarse labels as temporal ground truth [14, 15]. The resulting challenge is to use large recording-labeled collections for feature learning while retaining the temporal precision supplied by only a limited number of dense interval-annotated cohorts. To address these representation and supervision needs, we propose NOR-TL, a Normal-Reference Temporal Learning framework. NOR-TL fuses temporally aligned features from a frozen normal-pretrained encoder and a trainable AF encoder for local rhythm discrimination. Large recording-labeled datasets support both pretraining and downstream learning, with downstream supervision decoupled through a shared predictor: weak recording labels supervise pooled AF detection, while a limited number of dense interval annotations supervise sample-wise AF prediction. Our contributions are threefold:

(1) Normal-reference temporal learning. We introduce a dual-stream representation that combines a frozen normal-pretrained encoder with an adaptive AF encoder, providing stable normal-reference context together with task-specific AF features.

(2) Decoupled supervision across label granularities. We assign detection and temporal objectives according to annotation granularity, avoiding competing detection targets on densely annotated windows while retaining weak supervision during downstream learning.

(3) Comprehensive evaluation across temporal resolutions. We evaluate both window-level AF recognition and time-point AF detection on the publicly available and annotated AFDB, LTAfDB, and CPSC2021 datasets through comparisons with three locally adapted methods and controlled ablations of normal-reference conditioning and supervision routing.

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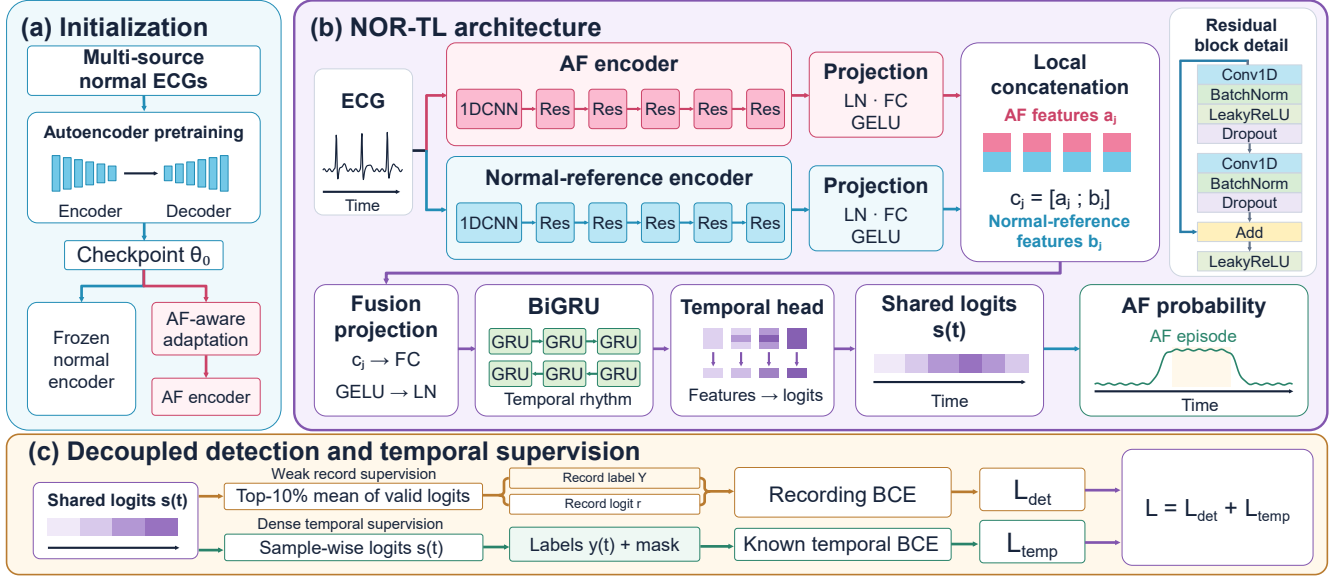


Fig. 1. Overview of NOR-TL. (a) Waveform reconstruction on normal ECGs and AF adaptation learn normal-reference and AF-specific representations. (b) The frozen normal encoder and trainable AF encoder process identical ECG inputs; their features are fused for temporal prediction. (c) Weak recording labels supervise pooled detection, and dense interval labels drive sample-wise AF prediction via the shared predictor.

2. METHOD

2.1. AF detection and model overview

NOR-TL processes single-channel ECGs in 10-s windows, each represented as $x_i \in \mathbb{R}^{2500}$. Training uses two annotation types: a recording label Y_i indicates AF presence, whereas a temporal label y_{it} identifies AF or non-AF at sample t . Temporal labels are obtained from annotated episode intervals; unknown portions and padding are excluded from their loss. The target follows the AF-family labels, including atrial fibrillation and atrial flutter.

As shown in Fig. 1(b), two encoders extract corresponding local features from the same ECG. Their fused sequence produces temporal AF probabilities p_{it} , which indicate AF likelihood at each sample. Pooling the underlying logits also gives a recording probability q_i , allowing recording and interval annotations to supervise the shared predictor at their respective resolutions.

2.2. Normal-reference temporal learning

We first pretrain a one-dimensional residual autoencoder on multi-source normal ECGs, using waveform reconstruction to establish its feature-extraction capability (Fig. 1(a)). The pretrained encoder initializes the normal-reference stream. A copy undergoes AF-specific adaptation with AF-positive, normal, and non-AF abnormal examples to initialize the AF stream (Section 3.2). The decoder is used only during pretraining.

Each encoder uses channel widths of 32, 64, 128, 256, 512, and 512, with kernel size five and successive downsampling factors of 2, 2, 2, 2, and 1. Channel reduction and adaptive pooling produce a 156×128 feature sequence. During downstream learning, the normal encoder remains frozen, while the AF encoder is updated with fixed batch-normalization running statistics.

Let E_A and E_N denote the AF and normal encoders. At each position j , independent trainable projections P_A and P_N , each comprising layer normalization, a linear layer, and Gaussian error linear

unit (GELU), produce

$$a_{ij} = P_A(E_A(x_i)_j), \quad b_{ij} = P_N(E_N(x_i)_j), \quad (1)$$

where $a_{ij}, b_{ij} \in \mathbb{R}^{128}$ and $j = 1, \dots, 156$. We concatenate the corresponding features and project them into a shared temporal representation:

$$u_{ij} = \text{LN}(\text{GELU}(W_f[a_{ij}; b_{ij}] + c_f)), \quad (2)$$

where $W_f \in \mathbb{R}^{256 \times 256}$ and $c_f \in \mathbb{R}^{256}$ are the fusion weights and bias. This fusion provides the temporal model with adaptive AF features and normal-trained context at each position.

A single-layer bidirectional gated recurrent unit (BiGRU) with 64 hidden units per direction processes the fused sequence. Layer normalization and a linear head produce local logits, followed by nearest-valid edge extension and linear interpolation to the 2,500-sample input grid. The resulting logits s_{it} yield $p_{it} = \text{sigmoid}(s_{it})$. Valid positions are packed for recurrent processing, while unknown annotation intervals retain their original temporal positions. Recurrent states are reset for each window, providing bidirectional context within that window.

2.3. Decoupled detection and temporal supervision

Let $\mathcal{W}$ denote weakly labeled recordings used for detection supervision and $\mathcal{D}$ denote densely annotated windows with at least one known temporal label. Both sets train the same temporal predictor (Fig. 1(c)). We use pooled logits for detection on $\mathcal{W}$ and sample-wise interval supervision on $\mathcal{D}$; throughout this section, BCE denotes binary cross-entropy computed from logits.

For weakly labeled recordings, top- k pooling aggregates temporal evidence into a recording-level detection score. Let $V_i = \{t : v_{it} = 1\}$ denote valid signal positions, excluding padding, and let K_i index the $k_i = \lceil 0.1|V_i| \rceil$ largest logits in V_i . The recording logit and probability are

$$r_i = \frac{1}{k_i} \sum_{t \in K_i} s_{it}, \quad q_i = \text{sigmoid}(r_i). \quad (3)$$

The pooled logit r_i is supervised by the recording label Y_i :

$$\mathcal{L}_{\text{det}} = \frac{1}{|\mathcal{W}|} \sum_{i \in \mathcal{W}} \text{BCE}(r_i, Y_i). \quad (4)$$

This objective supervises recording-level AF evidence, while the interval objective below supplies the temporal targets.

For a densely annotated window, $m_{it} = 1$ identifies samples that are both valid and temporally annotated. Let $M_i = \sum_t m_{it}$ be their number. Temporal supervision uses the interval labels y_{it} to penalize missed AF time and false-positive extension into non-AF time:

$$\mathcal{L}_{\text{temp}} = \frac{1}{|\mathcal{D}|} \sum_{i \in \mathcal{D}} \frac{1}{M_i} \sum_t m_{it} \text{BCE}(s_{it}, y_{it}). \quad (5)$$

The loss is averaged within each window and then across windows, so varying annotation coverage does not change the weight of a window. The two terms are combined with equal weights:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \mathcal{L}_{\text{temp}}. \quad (6)$$

Four-source recordings provide detection supervision, whereas interval-annotated AFDB windows provide temporal supervision.

The gradient directions explain why dense windows receive only interval supervision. For one dense positive window, consider an additional detection loss $\ell_{\text{rec}} = \text{BCE}(r_i, 1)$, and let ℓ_{temp} denote its mean temporal BCE. At a selected, known non-AF sample $t \in K_i$ with $y_{it} = 0$, the gradients are

$$\frac{\partial \ell_{\text{rec}}}{\partial s_{it}} = \frac{q_i - 1}{k_i} < 0, \quad \frac{\partial \ell_{\text{temp}}}{\partial s_{it}} = \frac{p_{it}}{M_i} > 0, \quad (7)$$

with the top- k selection held locally fixed. At a selected non-AF position, the positive detection objective increases the logit, whereas temporal supervision decreases it. To avoid imposing these competing targets on densely annotated windows, NOR-TL applies detection supervision to weakly labeled recordings and temporal supervision to dense windows. Both objectives update the shared trainable components, allowing recording-level data to strengthen AF discrimination while interval annotations supervise sample-wise predictions.

3. EXPERIMENTS

3.1. Datasets and preprocessing

To make full use of large-scale public ECG datasets, we use CODE15 [1], PTB-XL [9], CPSC2018 [16], and Chapman-Shaoxing [10] for pretraining and recording-level supervision. Lead I emulates the single-lead input setting of wearable and home monitoring. Screening retains usable recordings with AF-family, clean-normal, or non-AF abnormal labels. Table 1 reports the screened cohorts.

AFDB [8] provides dense supervision and held-out testing; the Long-Term AF Database (LTAfDB) [17] and the China Physiological Signal Challenge 2021 (CPSC2021) [18] are used exclusively for external testing. Long recordings are divided into 10-s windows. All windows from the same patient or original recording are kept in a single partition to prevent window-level leakage. Signals are calibrated in millivolts and resampled to 250 Hz.

3.2. Implementation and training

Encoder initialization used 40 epochs of normal ECG pretraining followed by 10 epochs of AF adaptation. NOR-TL was then trained

Table 1. Summary of datasets used in training and evaluation.

Dataset	Unit	Retained	Train	Val.	Test
CODE15	Recordings	78,962	56,346	10,048	12,568
PTB-XL	Recordings	9,481	7,167	1,243	1,071
CPSC2018	Recordings	5,019	3,792	671	556
Chapman-Shaoxing	Recordings	8,146	6,172	1,068	906
AFDB	Recordings	23	16	2	5
	Windows	84,310	58,892	7,013	18,405
LTAfDB	Recordings	84	—	—	84
	Windows	705,778	—	—	705,778
CPSC2021	Recordings	1,425	—	—	1,425
	Windows	173,537	—	—	173,537

for 10 epochs using AdamW, with learning rates of 10^{-5} for the AF encoder and 10^{-3} for new layers, weight decay 10^{-4} , and gradient clipping at five. Each batch contained 32 weakly and 32 densely labeled examples. Training was performed on an NVIDIA GeForce RTX 5070 Ti GPU. Model checkpoints were selected according to the mean validation recording and temporal AP. Decision thresholds were selected on the validation set and fixed for all test evaluations.

3.3. Results and discussion

3.3.1. Main performance comparisons

We compared NOR-TL with three publicly implemented methods: the waveform CRNN of Wen and Kang [19], SAFER [20], and the three-beat ResNet-Transformer [21]. All three were adapted to single-lead inputs and retrained using common data splits, following the publicly released code and the corresponding training procedures of the original methods. Additional references include MMA-RNN [22], evaluated on a separate patient-disjoint CPSC2021 split, and published 5-s AFDB classification results for HAN-ECG [23].

Table 2 shows that NOR-TL achieves the highest window-level and time-point Matthews correlation coefficient (MCC) across all three test cohorts. On AFDB, NOR-TL achieves the highest temporal MCC (0.465) and AUROC (85.3%). Evaluation on LTAfDB recordings further demonstrates leading performance across all four window-level metrics, with temporal MCC and AUROC reaching 0.837 and 98.2%, respectively. Although temporal sensitivity is lower than SAFER’s (85.6% versus 88.1%), higher specificity (98.1% versus 94.3%) yields a better MCC (0.837 versus 0.823). On CPSC2021, our largest external cohort, NOR-TL achieves a temporal MCC of 0.838, outperforming SAFER (0.735), while improving sensitivity from 73.4% to 85.5% and specificity from 95.9% to 96.5%. These gains indicate a higher proportion of AF time correctly labeled and fewer false-positive labels during non-AF periods. Collectively, these results show that NOR-TL provides a favorable balance between AF detection and non-AF rejection.

Beyond quantitative benchmark comparisons, Fig. 2 qualitatively illustrates sample-wise AF predictions on two representative CPSC2021 excerpts, covering challenging short-duration and long-duration AF episodes. For the brief AF segment, SAFER and CRNN completely miss the short AF episode, while 3-beat R-T detects only part of it and introduces false-positive predictions. For the longer episode, SAFER misses most of the AF interval, whereas CRNN and 3-beat R-T produce fragmented detections with intermittent missed segments. In contrast, NOR-TL captures the complete range of the challenging short AF episode and achieves substantially continuous,

Table 2. AF recognition and temporal labeling performance.

Model	Level	Sens.	Spec.	AUROC	MCC
AFDB					
CRNN [19]	W	75.1	52.3	69.7	0.281
	T	62.8	55.3	69.8	0.180
SAFER [20]	W	37.7	60.1	53.5	-0.022
	T	38.2	60.7	55.1	-0.011
3-beat R-T [21]	W	57.0	91.3	87.0	0.494
	T	86.6	45.7	76.3	0.359
NOR-TL	W	77.2	73.4	87.3	0.503
	T	65.3	81.6	85.3	0.465
LTAFDB					
CRNN [19]	W	87.0	94.4	96.2	0.811
	T	78.7	96.0	96.4	0.751
SAFER [20]	W	88.2	94.2	97.3	0.821
	T	88.1	94.3	97.4	0.823
3-beat R-T [21]	W	74.2	92.8	92.9	0.674
	T	90.2	81.7	92.9	0.724
NOR-TL	W	90.4	97.6	98.4	0.877
	T	85.6	98.1	98.2	0.837
CPSC2021					
CRNN [19]	W	72.7	95.4	94.0	0.720
	T	58.9	96.9	94.7	0.637
SAFER [20]	W	73.6	95.6	94.9	0.731
	T	73.4	95.9	95.4	0.735
3-beat R-T [21]	W	37.6	96.8	91.2	0.457
	T	71.7	87.9	90.5	0.604
NOR-TL	W	92.9	94.1	98.2	0.863
	T	85.5	96.5	98.0	0.838
<i>Independent / published protocols</i>					
MMA-RNN [†] [22]	T	81.6	92.4	93.1	0.735
HAN-ECG [‡] [23]	W	90.5	79.5	89.5	–

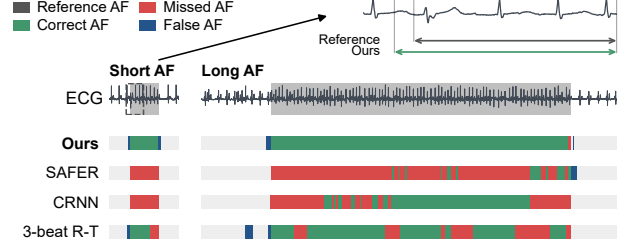
W: window recognition; T: time-point labeling. [†]MMA-RNN: re-evaluated following its patient-disjoint CPSC2021 setup. [‡]HAN-ECG: quoted from published 5-s AFDB classification results.

stable coverage over the long AF interval, with substantially fewer missed detections. The magnified ECG inset shows that NOR-TL detects the short-episode onset approximately 0.19 s before the manual annotation, illustrating close temporal alignment with the annotated rhythm transition.

3.3.2. Ablation studies

We conducted controlled ablations on AFDB to assess normal-reference conditioning and decoupled supervision for time-point AF detection. All variants follow the same training protocol, with changes restricted to the component under evaluation.

Normal-reference conditioning. Variants B and C demonstrate the contribution of the normal-reference design. Removing the reference encoder (B) reduced temporal MCC from 0.465 to 0.258 and AUROC from 85.3% to 74.7%, with decreases in both sensitivity and specificity. Replacing the normal-pretrained reference with an AF-pretrained encoder (C) increased sensitivity but reduced specificity to 60.0% and MCC to 0.404. The increased AF detection came with substantially more false positives in non-AF intervals. Together, the B/C comparisons highlight the role of normal-reference conditioning in suppressing false-positive AF predictions and improving the balance between AF detection and non-AF rejection.

**Fig. 2.** Qualitative comparison of sample-wise AF predictions. The inset shows an enlarged ECG segment around an annotated AF onset.**Table 3.** Ablation study on AFDB time-point AF detection.

Variant	Sens.	Spec.	AUROC	MCC
<i>Normal-reference conditioning</i>				
B: No reference encoder	58.6	67.5	74.7	0.258
C: AF-pretrained reference	79.5	60.0	68.1	0.404
<i>Decoupled supervision</i>				
D: + Dense detection BCE	40.4	84.4	78.0	0.268
E: W/o downstream weak BCE	27.7	56.5	51.3	-0.165
A: Full NOR-TL	65.3	81.6	85.3	0.465

Decoupled supervision. Variants D and E assess the contributions of decoupled supervision and continued use of weak recording labels during downstream learning. Adding recording-level BCE to densely annotated windows (D) increased specificity to 84.4% but reduced sensitivity from 65.3% to 40.4%, yielding lower AUROC (78.0%) and MCC (0.268). Removing downstream weak recording-level supervision (E), while retaining pretrained encoders and dense temporal supervision, severely degraded time-point detection performance: MCC fell to -0.165, sensitivity to 27.7%, and AUROC to 51.3%. Together, these results demonstrate the importance of assigning recording and interval labels to their respective objectives: separate temporal supervision improves AF coverage in dense windows, while continued weak supervision sustains AF discrimination beyond pretraining.

Overall, the full model achieved the highest temporal AUROC and MCC on AFDB, supporting the combined contributions of normal-reference temporal learning and decoupled supervision to time-point AF detection.

4. CONCLUSION

We proposed NOR-TL for single-lead PAF monitoring, combining normal-reference temporal learning with decoupled supervision. A normal-pretrained encoder provides stable reference features that are fused with those of an adaptive AF encoder at corresponding temporal positions. Weak recording labels supervise pooled AF detection, while dense interval labels supervise sample-wise predictions through a shared predictor. This design combines large-scale recording-labeled data with limited interval annotations for temporal AF learning. Across AFDB, LTAFDB, and CPSC2021, NOR-TL achieved the highest temporal MCC among the adapted baselines, with temporal AUROC and MCC reaching 98.0% and 0.838 on CPSC2021, respectively. Ablations supported the contributions of normal-reference conditioning and decoupled supervision. While the Lead-I ECG settings fit wearable devices, all evaluations in this study were conducted on clinical-grade recordings. Future work will further validate model robustness on real-world wearable ECG data.

5. COMPLIANCE WITH ETHICAL STANDARDS

This study involved secondary analysis of publicly available, de-identified ECG datasets used in accordance with their respective data-use conditions. No new human-subject data were collected. Ethical approval was not required for this secondary analysis.

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